sung claims that with these two technologies the sensor can instantaneous autofocus even in extreme lightning conditions.

Omnivision OVB0B. In January 2022, Omnivision launched its first 200MP camera sensor, which claims to have the smallest pixel measuring just $0.61\mu m \times 0.61\mu m$. For superfast autofocus, it employs 100% Quad Phase Detection (QPD), which also makes it the first 200MP sensor to offer this feature. Like Samsung's HP1, it also features a 4×4 pixel binning to provide an output of 12.5MP at 30fps with 3-exposure staggered HDR timing. The sensor supports CPHY, DPHY, and dual DOVDD for interfacing.

Gigajot Gj00422. Gigajot has released the Gj00422, a photon counting, Quanta Image Sensor (QIS). This is the first sensor capable of photon counting at room temperatures at full speed while achieving full high dynamic range. The Gj00422 is a 4MP sensor with square pixels measuring 2.2μm. Unlike the traditional CMOS sensor, where numerous photons are collected inside a pixel well and converted into electrons, QIS operates by detecting individual pixels in order to work in a poor-light environment with much higher fidelity than other types of CMOS sensors. It also features an on-chip temperature sensor and high intra-scene dynamic range. This sensor is suitable for astronomy, medical, industrial, defence, and other high-performance applications.

Future of imaging sensors

The camera technology has come a long way, and the advancement does not seem to be slowing down any time soon. The improvement in SSD devices means that higher resolution in smaller-size sensors is a safe bet to place.

Recently, researchers from Princeton University were able to fabricate a camera the size of a salt grain that can capture high-quality images, which means that the size of imaging sensors can be reduced to unperceivable sizes.

With the growth number of individual content creators, smartphone manufacturers are forced to improve the cameras in their devices. Many smartphone manufacturers are planning to use a bigger 2.5cm (one-inch) sensor, artificial intelligence, and high-quality lenses to improve camera capabilities. But only time will reveal how cameras improve in the coming years. EFY

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